

**Candidate Name:** Mayank Sethi

**Centre Number:** IN827

**Candidate Number:**2521

Insert your name, centre number and candidate number into the header above.

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Screenshots or program listings must be copied into appropriate cells in the following tables.

Examiners must be able to read the contents including any screenshots without the use of a magnifying glass. Answers that are not readable or missing will not be awarded any marks.

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## Question 1

### Part 1(a)

```
class node():  
    def __init__(self, data, next_node) -> None:  
        self.data = data  
        self.next_node = next_node
```

### Part 1(b)

```
linkedList:list[node] = [  
    node(1,1),  
    node(5,4),  
    node(6,7),  
    node(7,-1),  
    node(2,2),  
    node(0,6),  
    node(0,8),  
    node(56,3),  
    node(0,9),  
    node(0,-1),  
]
```

```
startPointer = 0  
emptyList = 5
```

### Part 1(c)(i)

```
def outputNodes(array:list[node], start_pointer:int) -> None:  
    current_pointer = start_pointer  
    if current_pointer != -1:
```

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```
print(array[current_pointer].data)
outputNodes(array=array, start_pointer=array[current_pointer].next_node)
```

### Part 1(c)(ii)

```
PS D:\Coding\CS-papers-solved> & C:\Python312\python.exe d:/Coding/CS-papers-solved/2021/mj-41/question1.py
Could not find platform independent libraries <prefix>
1
5
2
6
56
7
```

### Part 1(d)(i)

```
def addNode(array:list[node], start_pointer:int, empty_list_pointer:int) -> bool:
    value = int(input("What value to add to the linked list: "))

    # Free node operations
    if empty_list_pointer == -1:
        return False # no free list
    else:
        array[empty_list_pointer].data = value
        target_node = empty_list_pointer
        empty_list_pointer = array[empty_list_pointer].next_node

    # insertion
    if start_pointer == -1:
        return False # empty list
    current_node = start_pointer
    prev_node = -1
    while current_node != -1:
        prev_node = current_node
        current_node = array[current_node].next_node

    array[prev_node].next_node = target_node
    array[target_node].next_node = -1
    return True
```

### Part 1(d)(ii)

```
# d ii
result = addNode(linkedList, startPointer, emptyList)
if addNode:
    print("Element added successfully")
else:
```

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```
print("Element could not be added")
```

```
outputNodes(linkedList, startPointer)
```

### Part 1(d)(iii)

```
PS D:\Coding\CS-papers-solved> & C:\Python312\python.exe d:/Coding/CS-papers-solved/2021/mj-41/question1.py
Could not find platform independent libraries <prefix>
1
5
2
6
56
7
What value to add to the linked list: 5
Element added successfully
1
5
2
6
56
7
5
PS D:\Coding\CS-papers-solved> |
```

## Question 2

### Part 2(a)

```
# a
arrayData:list[int] = [10,5,6,7,1,12,13,15,21,8]
```

### Part 2(b)(i)

```
def linearSearch(target:int) -> bool:
    for i in range(len(arrayData)):
        if arrayData[i] == target:
            return True
    return False
```

### Part 2(b)(ii)

```
value = int(input("Enter the value that should be searched: "))
result = linearSearch(value)
if result:
    print(f"{value} was found")
else:
    print(f"{value} was NOT found")
```

### Part 2(b)(iii)

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```
PS D:\Coding\CS-papers-solved> & C:\Users\lenovo\AppData\Local\Programs\Python\Python311\python.exe d:/Coding/CS-papers-solved/2021/mj-41/question2.py
Enter the value that should be searched: 328957
328957 was NOT found
PS D:\Coding\CS-papers-solved> & C:\Users\lenovo\AppData\Local\Programs\Python\Python311\python.exe d:/Coding/CS-papers-solved/2021/mj-41/question2.py
Enter the value that should be searched: 10
10 was found
```

## Part 2(c)

```
def bubbleSort():
    temp:int
    for x in range(0, 9):
        for y in range(0,9-x):
            if arrayData[y] > arrayData[y+1]:
                temp = arrayData[y]
                arrayData[y] = arrayData[y+1]
                arrayData[y+1] = temp
```

## Question 3

### Part 3(a)

```
# a
class TreasureChest():
    def __init__(self, question:str, answer:str, points:int) -> None:
        # string self.__question
        # string self.__answer
        # integer self.__points
        self.__question: str = question
        self.__answer: str = answer
        self.__points: int = points
```

### Part 3(b)

```
# arrayTreasure ARRAY of TreasureChest
arrayTreasure:list[TreasureChest] = []
def readData() -> None:
    try:
        with open("./2021/mj-41/TreasureChestData.txt", 'r') as f:
            content = f.read()
            content = content.split("\n")
            count = 0
```

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```

    question = ""
    answer = ""
    points = 0
    for line in content:
        count +=1
        match count:
            case 1:question = line
            case 2: answer = line
            case 3:
                points = int(line)
                arrayTreasure.append(TreasureChest(question, answer, points))

        if count == 3:
            count = 0
    except FileNotFoundError:
        print("File not able to be found")

```

### **Part 3(c)(i)**

```

def getQuestion(self) -> str:
    return(self.__question)

```

### **Part 3(c)(ii)**

```

def checkAnswer(self, answer:str) -> bool:
    if answer == self.__answer:
        return True
    return False

```

### **Part 3(c)(iii)**

```

def getPoints(self, attempt_count:int) -> int:
    if attempt_count > 4 or attempt_count < 1:
        return 0
    return self.__points//attempt_count

```

### **Part 3(c)(iv)**

```

readData()

question_number = -1
while question_number < 1 or question_number > 5:
    question_number = int(input("Enter a number between 1 to 5 to get a question: "))

Treasure = arrayTreasure[question_number-1]

print(Treasure.getQuestion())

correct = False
attempts = 0
while not correct:

```

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```
answer = input("answer: ")
correct = Treasure.checkAnswer(answer)
attempts = attempts + 1
```

```
points = Treasure.getPoints(attempts)
```

```
print(f"You have been awarded with {points} points!")
```

### Part 3(c)(v)

```
PS D:\Coding\CS-papers-solved> & C:\Users\lenovo\AppData\Local\Programs\Python\Python311\python.exe d:/Coding/CS-papers-solved/2021/mj-41/question3.py
21/mj-41/question3.py
PS D:\Coding\CS-papers-solved> & C:\Users\lenovo\AppData\Local\Programs\Python\Python311\python.exe d:/Coding/CS-papers-solved/2021/mj-41/question3.py
Enter a number between 1 to 5 to get a question: 1
2*2
● answer: 4
You have been awarded with 10 points!
PS D:\Coding\CS-papers-solved> & C:\Users\lenovo\AppData\Local\Programs\Python\Python311\python.exe d:/Coding/CS-papers-solved/2021/mj-41/question3.py
Enter a number between 1 to 5 to get a question: 5
3000+4000
answer: 12
○ answer: 7000
You have been awarded with 9 points!
```